**National Research University Higher School of Economics**

**Faculty of Computer Science**

**Bachelor Programme "Data Science and Business Analytics"**

# Course Project

## Development of an Alice Voice Assistant Skill for International Applicants of HSE

**Student:**
Kovalenko Mikhail Alexandrovich
Year: 2
**Supervisor:**
Scientific supervisor, Andreeva Darya Alexandrovna

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# Introduction

Voice assistants such as Alice (Yandex), Marusya (VK), and Salute (Sber) have become an important component of the modern artificial intelligence ecosystem, enabling users to access information through natural voice interaction. In the Russian Federation, Alice occupies a leading position among voice assistants due to its strong support for the Russian language, high-quality speech recognition, and deep integration with the Yandex service infrastructure. The Yandex.Dialogs platform enables the development of so-called "skills," which are server-side applications integrated via webhook that process user requests and return responses within the voice assistant ecosystem.

A particularly important category within the university admission system is international applicants. Unlike domestic applicants, they face additional challenges, including language barriers, time zone differences, and the need to navigate the specific administrative and educational systems of the Russian Federation. Admission-related information is often distributed across multiple official documents and web pages, making it difficult for international applicants to independently interpret and structure the required information.

The Higher School of Economics annually attracts a significant number of international applicants, which increases the workload of the admissions office, especially during peak admission periods. Traditional communication channels—such as the official website, email correspondence, and phone consultations—do not always provide timely and convenient responses for users located outside the Russian Federation.

Under these conditions, the development of a specialized voice assistant skill for Alice, targeted specifically at international applicants of HSE, represents a relevant and practical solution. The use of a Retrieval-Augmented Generation (RAG) architecture, which retrieves relevant fragments of official documents before generating a response, allows the system to preserve factual consistency and minimize the risk of hallucinated or outdated information by grounding answers in official university sources.

# Project Goal

The goal of this project is to develop a voice assistant skill titled *"HSE International Applicant"* for the Alice voice assistant, which provides international applicants of the Higher School of Economics with up-to-date and reliable information about the university admission process.

# Project Objectives

To achieve the stated goal, the following objectives were defined:

1. Analyze the informational needs of international applicants and identify typical inquiry scenarios.
2. Design a system architecture based on the RAG approach, including a retrieval module, knowledge index, and response generation module.
3. Construct and structure a knowledge base derived from official HSE admission documents and web pages.
4. Perform data preprocessing, including text cleaning, chunk segmentation, and metadata annotation.
5. Implement vector-based document indexing using embedding models.
6. Conduct experimental comparison of different embedding models and evaluate their impact on retrieval quality.
7. Investigate the influence of retrieval parameters (e.g., top-k) on contextual completeness and relevance.
8. Compare the effectiveness of vector-based retrieval and keyword-based search methods.
9. Develop fallback strategies for cases of low-confidence retrieval or missing relevant documents.
10. Evaluate system responses in terms of factual grounding, completeness, robustness against hallucinations, and response latency.
11. Conduct system testing and formulate recommendations for further improvement and scalability.

# Literature Review

Voice assistants represent a rapidly developing field within artificial intelligence, combining speech recognition, natural language processing, and speech synthesis technologies. In the educational domain, such systems are increasingly used to automate responses to frequently asked questions and to optimize communication between users and institutions.

With the advancement of large language models (LLMs), the issue of hallucination—generation of factually incorrect or fabricated information—has become a critical challenge. In domains that require strict adherence to official regulations, such as university admissions, such errors are unacceptable. To address this problem, the Retrieval-Augmented Generation (RAG) architecture was proposed. RAG combines a document retrieval component with a generative language model, where the model produces responses conditioned on retrieved evidence from an external knowledge base.

The RAG approach constrains generation to verified document fragments, thereby significantly improving factual grounding and reducing the likelihood of hallucinated content. Recent extensions of RAG include hybrid retrieval (combining vector similarity and keyword filtering), query expansion techniques, and confidence-based fallback strategies. These mechanisms are particularly relevant in high-stakes informational domains, including legal, medical, and academic admission systems.

Within the Russian technological ecosystem, the Yandex.Dialogs platform provides a ready-to-use infrastructure for developing voice assistant skills. Leveraging an existing platform substantially reduces technical complexity compared to building a standalone voice assistant, which would require implementing a complete technology stack (Speech-to-Text, NLP, and Text-to-Speech) as well as dedicated cloud infrastructure.

An analysis of existing informational tools for HSE applicants reveals the absence of a specialized voice-based solution grounded in official university documents. This confirms the relevance of developing a voice assistant skill that integrates Yandex.Dialogs infrastructure with a RAG-based architecture to provide accurate, grounded, and accessible informational support for international applicants of the Higher School of Economics.

# 1 Experimental Evaluation

## 1.1 Dataset

The system was evaluated on a manually constructed gold dataset consisting of 50 questions related to admission procedures for international applicants to HSE University.

Each dataset entry contains:

- the user question,
- the identifier of the relevant source document,
- the reference answer (used for manual validation).

The document corpus includes 10 official admission-related documents in `.docx` format. After preprocessing and chunking, the corpus produced 314 text chunks used for indexing.

## 1.2 Retrieval Configuration

ChromaDB was used as the vector storage and retrieval engine. Dense embeddings were generated using the `sentence-transformers/all-MiniLM-L6-v2` model.

The selected chunking configuration:

- Chunk size: 900 characters
- Overlap: 150 characters

For each query, the retriever returns the top-$k$ most relevant chunks, where $k \in \{1, 3, 5\}$.

### 1.3 Evaluation Metrics

We use standard retrieval metrics:

- **Hit@k**: equals 1 if the correct document appears among the top-$k$ retrieved results.
- **MRR@k (Mean Reciprocal Rank)**: evaluates how high the correct document appears in the ranking.

Mean Reciprocal Rank is defined as:

$$MRR = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{rank_i}$$

where $rank_i$ denotes the position of the correct document for query $i$.

### 1.4 Baseline Retrieval Performance

The baseline configuration (MiniLM embeddings, chunk size 900/150) was evaluated on 50 gold questions.

| | Hit@1 | Hit@3 | Hit@5 | MRR@1 | MRR@3 | MRR@5 |
|---|---|---|---|---|---|---|
| Baseline (900/150) | 0.20 | 0.30 | 0.30 | 0.20 | 0.2467 | 0.2467 |

Table 1: Baseline dense retrieval performance

### 1.5 Impact of Chunk Size

| Chunk Size / Overlap | Hit@1 | Hit@3 | Hit@5 | MRR@1 | MRR@3 | MRR@5 |
|---|---|---|---|---|---|---|
| 600 / 100 | 0.14 | 0.22 | 0.32 | 0.14 | 0.1733 | 0.1963 |
| 900 / 150 | **0.20** | **0.30** | 0.30 | **0.20** | **0.2467** | **0.2467** |
| 1200 / 200 | 0.12 | 0.20 | 0.20 | 0.12 | 0.1533 | 0.1803 |
| 1500 / 250 | 0.12 | 0.14 | 0.24 | 0.12 | 0.1300 | 0.1500 |

Table 2: Effect of chunk size on retrieval quality

## 1.6 Discussion

The results indicate that medium-sized chunks (900/150) provide the best balance between contextual completeness and ranking precision.

Smaller chunks slightly improve Hit@5 but reduce ranking quality (lower MRR), indicating context fragmentation. Larger chunks decrease retrieval accuracy due to semantic dilution.

## 1.7 Conclusion of Experimental Study

The implemented dense retrieval system demonstrates baseline performance for admission-related queries:

- Hit@1 = 0.20
- Hit@3 = 0.30
- MRR@5 = 0.2467

Future improvements may include testing stronger multilingual embedding models, hybrid retrieval approaches (BM25 + dense retrieval), and reranking techniques to enhance ranking precision.

**Project Repository:**
https://github.com/MikulaKula/Alice-applicants

# 2 Experiments and Evaluation

## 2.1 Evaluation Methodology

To evaluate the retrieval quality of the RAG pipeline, a gold evaluation dataset consisting of 50 question-answer pairs was created manually based on official HSE documents for international students. Each record in the dataset contains:

- a user question,
- the reference answer,
- the source document,
- the supporting fragment,
- thematic category,
- difficulty level.

The retrieval system was evaluated using the following metrics:

- **Hit@1**,
- **Hit@3**,
- **Hit@5**,
- **MRR@5** (Mean Reciprocal Rank).

Hit@k measures whether at least one relevant chunk appears among the top-k retrieved chunks. MRR evaluates how highly the first relevant chunk is ranked.

The baseline configuration used:

- ChromaDB vector database,
- dense retrieval over document chunks,
- fixed-size chunking,
- retrieval-only pipeline without reranking.

## 2.2 Preprocessing Experiments

Several preprocessing strategies were evaluated in order to determine how text normalization affects retrieval quality. The following preprocessing methods were tested:

- raw text without preprocessing,
- basic preprocessing,
- strict preprocessing.

Table 3: Preprocessing Experiment Results

| Method | Hit@1 | Hit@3 | Hit@5 | MRR@5 |
|---|---|---|---|---|
| Raw | 0.20 | 0.30 | 0.30 | 0.2467 |
| Basic | 0.22 | 0.30 | 0.30 | 0.2567 |
| Strict | 0.22 | 0.28 | 0.30 | 0.2550 |

The results demonstrate that moderate preprocessing slightly improves retrieval quality. However, overly aggressive preprocessing may remove useful contextual information from official university documents. The best overall performance was achieved using the basic preprocessing strategy.

## 2.3 Chunking Experiments

Chunking strategy significantly affects retrieval quality in RAG systems. Several chunk sizes and chunking methods were evaluated. The following configurations were tested:

- fixed chunking with overlap,
- paragraph-based chunking.

Table 4: Chunking Experiment Results

| Method | Hit@1 | Hit@3 | Hit@5 | MRR@5 |
|---|---|---|---|---|
| Fixed 600/100 | 0.14 | 0.22 | 0.30 | 0.1957 |
| Fixed 900/150 | 0.20 | 0.30 | 0.30 | 0.2450 |
| Fixed 1200/200 | 0.12 | 0.20 | 0.32 | 0.1803 |
| Fixed 1500/250 | 0.12 | 0.14 | 0.24 | 0.1500 |
| Paragraph 900/0 | 0.10 | 0.34 | 0.48 | 0.2343 |

The experiments showed that small chunks lose semantic context, while excessively large chunks reduce retrieval precision. The fixed chunk size of 900 characters with 150 overlap provided the most balanced performance overall.

Paragraph-based chunking achieved the best Hit@5 score because larger logical sections more frequently contained the correct answer. However, this method reduced Hit@1 precision.

## 2.4 Reranking Experiments

To improve retrieval quality, an additional lexical reranking stage was tested. The reranker re-ordered retrieved chunks using lexical similarity scores after dense retrieval.

Table 5: Reranking Experiment Results

| Method | Hit@1 | Hit@3 | Hit@5 | MRR@5 |
|---|---|---|---|---|
| Retrieval only | 0.20 | 0.28 | 0.30 | 0.2450 |
| Retrieval + lexical reranker | 0.20 | 0.28 | 0.30 | 0.2417 |

The lexical reranker did not improve retrieval quality. The results remained nearly identical to the baseline retrieval-only approach. One possible explanation is that official university documents contain highly formalized vocabulary and repetitive terminology, which are already captured effectively by dense retrieval methods.

## 2.5 Generation Evaluation

The generation component was evaluated manually. For selected questions from the gold dataset, answers were generated based on retrieved chunks and then evaluated according to the following criteria:

- factual correctness,
- relevance,
- completeness,
- absence of hallucinations,
- consistency with official HSE information.

Manual evaluation was selected because factual accuracy is critically important for university-related information systems. Incorrect or hallucinated answers may negatively affect international students during their adaptation process.

The evaluation demonstrated that retrieval quality strongly influences generation quality. Most generation errors were caused by incomplete or partially irrelevant retrieved chunks rather than by the generation process itself.

## 2.6 Discussion of Results

The experiments demonstrate that retrieval quality in educational RAG systems strongly depends on preprocessing and chunking strategies.

Moderate preprocessing improves retrieval stability, while excessive preprocessing may reduce contextual richness. Chunk size selection also plays a critical role. Medium-sized chunks achieved the best balance between semantic completeness and retrieval precision.

Reranking provided limited benefit in the current setup because the dataset consists primarily of structured official documents with repetitive terminology.

Overall, the developed RAG pipeline successfully retrieves relevant information from official HSE documents and can support international students in solving organizational and academic adaptation tasks.

## 2.7 Agentic RAG Experiment

An additional experiment was conducted using a simplified Agentic RAG approach. Instead of performing only direct retrieval, the system first selected an action depending on the user query type.

The Agentic RAG approach was selected instead of GraphRAG because the project domain mainly consists of procedural university documents, regulations, and FAQ-style information rather than highly interconnected entities and relationships. For this type of knowledge base, the main challenge is not graph traversal, but correct query routing, ambiguity handling, and filtering of irrelevant requests.

In addition, Agentic RAG is computationally simpler and easier to integrate into a lightweight university assistant system. The approach allows the system to dynamically decide whether retrieval is necessary, whether clarification should be requested, or whether the query should be rejected as off-topic. This behavior is especially important for real student interactions, where users often submit incomplete or unrelated questions.

The implemented agent supported several possible actions:

- `retrieve_answer` — perform standard retrieval from the vector database;
- `ask_clarification` — request clarification for ambiguous queries;
- `refuse_offtopic` — reject queries unrelated to the university domain.

Several test queries were used to evaluate the routing logic.

| **User Query** | **Selected Action** |
|---|---|
| What documents do I need for migration registration? | retrieve_answer |
| Dormitory | ask_clarification |
| What is the weather today? | refuse_offtopic |
| Where can I find the contact email? | refuse_offtopic |

Table 6: Agentic RAG routing behavior

The experiment demonstrated that the agent-based routing mechanism can improve robustness of the system by filtering irrelevant requests and handling ambiguous queries before retrieval. Compared to a standard RAG pipeline, the Agentic RAG approach provides an additional decision-making layer that improves interaction control and reduces unnecessary retrieval operations.